



VAIBHAV KATENDRA



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	6.92
2024	B.Tech in Information Technology	Bhilai Institute of Technology Durg	8.53
2019	XII: Central Board of Secondary Education	Jawahar Navodaya Vidyalaya Durg	88.2%
2017	X: Central Board of Secondary Education	Jawahar Navodaya Vidyalaya Durg	10.0

IIT DELHI THESIS

Title: Adversarial AI Attacks and Defenses in Intrusion Detection System (IDS) for IoT

Supervisor: Prof. Vireshwar Kumar

Description: Developing an **adversarial AI-driven IDS** for **IoT networks** by identifying exploitable attack surfaces through **fuzzing** and applying **adversarial machine learning** to design robust, low-false-alarm defenses. The framework is being validated on simulated networks and real-world IoT devices to detect **advanced AI-assisted cyber-attacks** with high accuracy, aiming to establish a systematic methodology for **IoT security**.

PROJECTS

- **Kernel-Level Systems Programming Projects – xv6 (Prof. Smruti Ranjan Sarangi)** (January, 2024 - April, 2024)
 - Enhanced **xv6 shell** with login authentication, process history, syscall block/unblock, and file permission support.
 - Implemented **signal handling** (terminate/suspend/resume/custom) and developed a **priority-boosting scheduler** with profiling of turnaround, waiting, and response times.
 - Built **memory management** extensions including per-process memory usage reporting, page swapping with adaptive replacement, and page-fault handling.
- **Advanced Networking Projects (Prof. Tarun Mangla)** (August, 2024 - November, 2024) :
 - Implemented **reliable UDP** with TCP-like features (ACKs, retransmissions, Reno & CUBIC congestion control).
 - Built **SDN controllers** (Learning Switch, Spanning Tree, Shortest Path, Congestion-Aware Routing) using **Ryu/Mininet**.
 - Designed **multithreaded** TCP client-server with FIFO/RR scheduling and collision protocols (Slotted Aloha, BEB).
- **Mini Web Browser (Prof. Smruti Ranjan Sarangi)** (October, 2024 - November, 2024) :
 - Developed a lightweight **multi-tab browser** in C++ with **multi-process/threaded** architecture enabling concurrent fetching, parsing, and rendering of HTML pages; tested on local HTML pages.
 - Built a custom **HTML parser** to construct a **DOM tree** and rendered content using low-level Qt widgets and QPainter, avoiding any use of built-in web engines; integrated **HTTP fetching** via libcurl with local caching.
- **Metal Slug Game (Prof. Smruti Ranjan Sarangi)** (September, 2024 - October, 2024) :
 - Developed a 2D run-and-gun **multiplayer** platformer in Unity3D with smooth controls, responsive UI, real-time physics, progressive levels, collision handling, **health/damage** systems, enemies, collectibles, sound effects, and power-ups.
- **Latex2Markdown Converter (Prof. Smruti Ranjan Sarangi)** (July, 2024 - August, 2024) :
 - Developed a LaTeX-to-Markdown converter that lexes, parses, and generates an **AST** which is able to handle sections, text formatting, lists, **tables with alignment**, code blocks, hyperlinks, **images**, and **comments**.
- **ShareWheels (BIT Durg)** (July, 2023 - December, 2023) :
 - Developed an Android carpooling app in **Flutter** with Firebase authentication, Google Maps integration, real-time GPS, **integrated payment**, ride search/offer, ride scheduling, and history features.
- **Additional project repositories and reports on [GitHub](#)**

TECHNICAL SKILLS

• C | C++ | Python | Java | Assembly (x86-64) | Bash | Linux | xv6 | Git | GDB | Make | Unity3D | Android | Flutter

SCHOLASTIC ACHIEVEMENTS

- Secured **AIR-370** in **GATE-Computer Science** 2024.
- Secured **AIR-997** in **GATE-Data Science & Artificial Intelligence** 2024.
- Achieved a perfect score (**100/100**) in **COP701 Software Systems Laboratory**, **IIT Delhi**.
- Represented **Chhattisgarh** at the **National Camp** of Vidyarthi Vigyan Manthan, hosted at **IIT Delhi**.

TEACHING ASSISTANTSHIP

- **COL106:** Data Structures and Algorithms (Spring 2025) | • **COL351:** Analysis and Design of Algorithms (Fall 2024)

Disclaimer: All the information on this page is entered by student.



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IIT COURSE

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COURSES DONE

Advanced Data Structures, Computer Networks, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Resource Management In Computer Systems, Networks & System Security, Minor Project